

AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 10.4: Case Study 3 — ES Fellows Dataset

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Topic 10.4 Agenda

1. **Case Study 3** — ES Fellows dataset construction with agentic AI

Case Study 3

A provenance-aware public-web dataset construction workflow

Why This Third Case Is Different

- the source material is public and semi-structured
- metadata quality is uneven across people and institutions
- missingness is part of the result, not a nuisance to hide
- the workflow must control sources, provenance, and coding rules

This is not a web-scraping stunt. It is a data-construction problem with explicit source restrictions.

Research Question for the Fellows Dataset

Question: What can we learn about career stage, election timing, and field composition among current Econometric Society fellows using only official and institutional sources?

- it is a useful dataset about the economics profession itself
- it exposes a new research pattern: roster seed + metadata enrichment + coverage reporting
- it forces us to separate current biological age, years since PhD, and age at fellow election
- it forces students to confront missingness, source quality, and coding judgment

Why a Naive Web-Chat Workflow Still Fails Here

1. **Roster control is weak.** You need a canonical seed list and fixed batches.
2. **Source discipline is weak.** You need a rule that excludes Wikipedia and open-web guesswork.
3. **State tracking is weak.** You need persistent files recording what was found and what stayed missing.
4. **Auditability is weak.** You need field-level source URLs, not just narrative answers.

Precise contrast: modern web chat can browse and summarize, but this assignment needs a project harness with files, batching, and provenance rules.

Official Seed Roster as of April 14, 2026

- canonical source: the official Econometric Society current fellows page
- parsed roster size: 882 fellows
- official CV links on the seed page: 64
- usable election years in the current page: 880; two entries parse as 0000
- batching rule: 18 batches of 50 fellows
- seed fields: name, affiliation, election year, profile URL, CV URL

Interpretation:

- the roster is large enough that the workflow must be batched
- CV coverage is sparse enough that missingness will remain visible
- the current roster is not the same thing as a confirmed-living roster

Batch Workflow and Project Structure

Project files

- seed roster CSV
- batch assignment file
- normalized batch CSV
- analysis script
- plots and memo

Workflow

1. parse the official roster
2. assign a batch of 50
3. fill official/institutional metadata
4. run analysis
5. write findings and limitations

This is the public-web analog of 'CLAUDE.md': the project files hold the rules and state across sessions.

Schema and Provenance Rules

Field	Rule
name, affiliation, election_year	seed roster only
bio_url, cv_url	official or institutional only
phd_year, re-search_fields_raw	record source URL for each field
normalized field bucket	map raw text into the lecture taxonomy
coverage_status	say what remains missing
notes	document coding judgment or ambiguity

Credibility requires traceable provenance for each field.

Missingness, Gender, and Living-Status Caution

- **Birth date:** leave missing unless the official source states it
- **PhD year:** use only if the source states the degree year directly
- **Gender:** code only from explicit pronouns or explicit gender language
- **No inference:** not from names, photos, or country
- **Living status:** do not infer from old election year alone; for lecture purposes, use a 90-year cap only as a conservative sensitivity rule

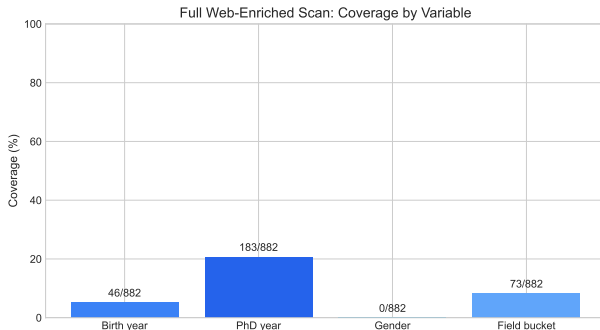
This is deliberate. The assignment is designed to teach credible coding rules, not aggressive completion.

Instructor Full Scan: What the Agentic Workflow Added

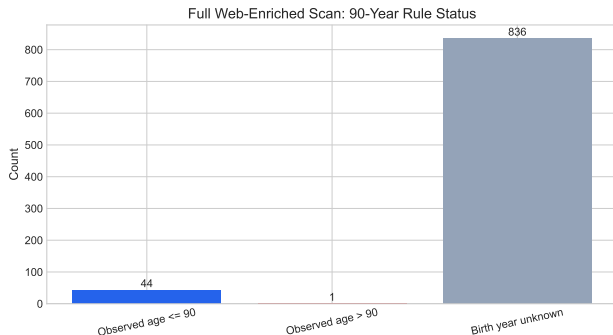
Metric	Official → Web
Local CV docs	62 → 316
Birth year	11 → 46
PhD year	47 → 183
Field bucket	15 → 73

Main implication:

- the agent scaled discovery and parsing well
- PhD-year coverage improved materially
- birth-year coverage stayed thin



The 90-Year Rule Does Not Solve the Main Problem

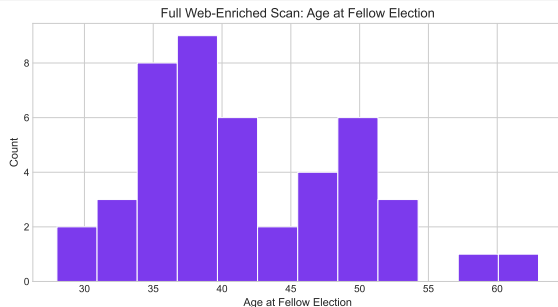
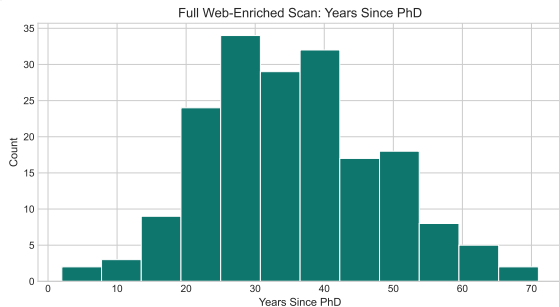


- observed age ≤ 90 : 44 fellows
- observed age > 90 : 1 fellow
- birth year unknown: 836 fellows

Takeaway:

- the problem is missing birth year, not a handful of extreme ages
- do not treat current biological age as the main variable
- use age at election or years since PhD instead

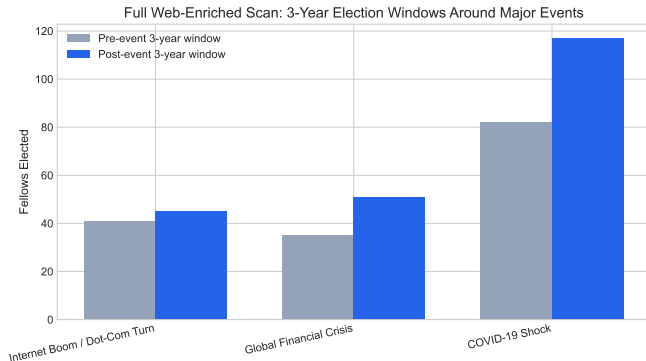
Career Stage: Use Years Since PhD and Age at Election



Main descriptive findings:

- median years since PhD: 35
- median age at fellow election: 40
- median years from PhD to election: 17
- age at election is cleaner than current age because it does not depend on fellows being alive today
- this is exactly why agent-generated datasets still need variable-design judgment

Election Counts Around Major Economics Events



3-year post-event windows

- Internet turn (2000): 41 → 45
- GFC (2008): 35 → 51
- COVID era (2020): 82 → 117

Interpretation:

- there is no obvious post-dot-com collapse
- counts rise after 2008 and surge after 2020
- the 2020–2024 jump likely reflects Society-side policy or catch-up effects
- use this as descriptive timing evidence, not a causal event study

What Agentic AI Actually Did Here

- discovered and cached candidate homepages at scale, then kept the search state on disk
- parsed 527 institutional profile pages and downloaded 316 CV or CV-like documents
- normalized fields, tracked source URLs, and reran the analysis after rule changes
- surfaced a negative result clearly: even a large search pass does not rescue current biological age

What stayed human:

- define source restrictions and coding rules
- separate current age, years since PhD, and age at election
- treat event patterns as descriptive rather than causal
- decide when missingness is too severe for a claim

Exercise B: Full-Roster Fellows Assignment

- start from the official seed roster and take one batch of 50 fellows
- use only official and institutional sources
- produce a normalized CSV, four figures, and a short memo
- keep missing values visible and documented
- use age only if birth-year coverage reaches the lecture threshold; otherwise use years since PhD or age at election
- explain clearly what the agent automated and what judgment you supplied

This case is not about scraping harder. It is about designing a trustworthy public-web research pipeline.